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EQUAL-TIME TEMPERATURE CORRELATORS OF THE ONE-DIMENSIONAL HEISENBERG XY CHAIN

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For equal-time temperature correlators of the anisotropic Heisenberg XY chain, representations are obtained in the form of determinants of $M \times M$ matrices. These representations are simple deformations of the answers for the isotropic XXO chain. In the thermodynamic limit, the correlators are expressed in terms of the Fredholm determinants of linear integral operators. Bibliography: 30 *titles.*

INTRODUCTION

The description of the correlation functions for the models solved by means of the Bethe ansatz is based on the representation of the correlators as the Fredholm determinants of linear integral operators. For the simplest two-point equal-time correlators of the one-dimensional model of impenetrable bosons, such representations were first obtained in [1, 2]. Later, they were generalized to the case of time-dependent correlators for the models that are the "free-fermion point" of the models solved by means of the Bethe ansatz (impenetrable bosons [3] and isotropic XXO Heisenberg chain [4, 5]) and also to the case of finite interaction ([6, 7] for the one-dimensional Bose-gas and [8] for XXZ chain). Such representations allow us to obtain classical integrable equations for correlators which can be used, in particular, for calculating the long time and large distance asymptotic series for the correlation functions [10, 12, 14, 15].

It has long been realized [16–18] that the language of the classical differential equations is very natural for the description of the correlation functions of quantum integrable models. The recent progress made in this direction based on the approach elaborated in [10–15] is described in detail in the book [19] for the example of impenetrable bosons. The heart of this approach is consideration of the Fredholm determinant of the linear operator appearing in the representation of the correlators as a τ-function for a new system of classical integrable equations (see also [9]). These linear operators, the so-called "integrable" integral operators (see [19, 20]), must be of a very special form.

In the present paper, we obtain some determinant representations for equal-time temperature correlators of the anisotropic Heisenberg XY chain both for a finite lattice and in the thermodynamic limit. This model was introduced and studied for the first time in [21]. Later it was studied by many authors (see [22–24] and the references therein).

To compute the correlation functions, we use a modification of the approach proposed in [25] and based on the use of the integration over the Grassmann variables and the corresponding coherent states. This is the main point in our paper. However, we do not use the functional integral as is done in [25]. The use of the coherent states simplifies the calculations and leads automatically to the answers in the required form (see, e.g., Appendix B, where we reproduce the results from [4, 5] for the XXO chain by our method).

The representations for the equal-time temperature correlators of the anisotropic XY chain in the form of determinants of $M \times M$ matrices obtained in this paper are, on one hand, direct generalizations of the representations for the isotropic case [4, 5, 26] and differ from them only in replacing the "Fermi" (or "Bose") weight in the kernels of integral operators by a weight depending on the anisotropy parameter. On the other hand, these representations generalize the results for the anisotropic chain obtained in [27] for zero temperature. The "integrability" of the integral operators appearing in the thermodynamic limit is obvious.

It should be noted that the isotropic XXO chain is the "free-fermion point" for the XXZ Heisenberg chain, which is a model solved by means of the standard Bethe ansatz. The anisotropic XY chain is the "free-fermion line" for the XYZ chain, which is a model where the usual Bethe ansatz does not work.

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Therefore, it is very interesting from our point of view that the answers for the equal-time correlators have the form of a simple deformation of the answers for the isotropic case. We hope that the corresponding description for the time-dependent correlation functions will be given in a forthcoming paper.

The present paper is organized as follows.

In the first section, we describe the model and present the basic facts about the diagonalization of the Hamiltonian by the Bogolyubov transformation. Here we mainly follow the classical papers [21, 23]. In Sec. 2, we introduce the coherent states for the isotropic XXO model and calculate the matrix elements of some operators between these states. In Sec. 3, we describe the coherent states for the anisotropic XY chain and consider the association between these states and the coherent states of the isotropic chain. In Sec. 4, the simplest correlator is calculated. In Sec. 5, we calculate the local spin correlator for the finite anisotropic lattice. In Sec. 6, we obtain the results in the thermodynamic limit. In Appendix A, the basic facts about Grassmannian coherent states are given. In Appendix B, the proof of the results for the isotropic case is presented.

1. XY Heisenberg chain

The Hamiltonian of the XY spin chain describing the interaction between the nearest neighbors (spins 1/2 located at the sites of a one-dimensional periodic lattice in a constant magnetic field h) can be represented in the form

$$\mathbf{H} = \mathbf{H}_0 + \gamma \mathbf{H}_1 - h\mathbf{S}^z, \tag{1.1}$$

where

$$\mathbf{H}_0 = -\frac{1}{2}\sum_{m=1}^{M} \left(\sigma_m^+\sigma_{m+1}^- + \sigma_m^-\sigma_{m+1}^+\right), \tag{1.2}$$

$$\mathbf{H}_1 = -\frac{1}{2}\sum_{m=1}^{M} \left(\sigma_m^+\sigma_{m+1}^+ + \sigma_m^-\sigma_{m+1}^-\right), \tag{1.3}$$

and third component of the total spin has the form

$$\mathbf{S}^z = \frac{1}{2}\sum_{m=1}^{M} \sigma_m^z. \tag{1.4}$$

We assume that the total number M of sites is even. The Pauli matrices are defined as usual,

$$[\sigma_m^\alpha, \sigma_n^\beta] = 2i\,\delta_{mn}\,\epsilon^{\alpha\beta\gamma}\,\sigma^\gamma \quad (\alpha,\beta,\gamma = x,y,z), \quad \sigma_m^\pm = (1/2)(\sigma_m^x \pm i\sigma_m^y), \tag{1.5}$$

with the periodic boundary conditions

$$\sigma_{M+1}^\alpha \equiv \sigma_1^\alpha. \tag{1.6}$$

Owing to the symmetries of the Hamiltonian, the sign of the magnetic field (as well as the sign of the Hamiltonian) is not essential. We assume that $h \geq 0$.

The Jordan–Wigner transformation

$$a_m = \exp\left\{i\pi\mathbf{Q}(m-1)\right\}\sigma_m^+, \quad a_m^+ = \sigma_m^- \exp\left\{-i\pi\mathbf{Q}(m-1)\right\} \tag{1.7}$$

introduces the canonical fermion fields a_m and a_m^+ on the lattice,

$$\left[a_m, a_n\right]_+ \equiv a_m\,a_n + a_n\,a_m = 0, \quad \left[a_m^+, a_n^+\right]_+ = 0. \quad \left[a_m\,a_n^+\right]_+ = \delta_{mn}. \tag{1.8}$$

Here $\mathbf{Q}(m)$ is the operator of the number of particles on the first m sites of the lattice,

$$\mathbf{Q}(m) = \sum_{j=1}^{m} q_j, \tag{1.9}$$

where q_m is the operator of the number of particles at the site m,

$$q_m = a_m^+ a_m = \sigma_m^- \sigma_m^+ = \frac{1}{2}(1 - \sigma_m^z). \tag{1.10}$$

The operator of the total number of particles

$$\mathbf{N} = \mathbf{Q}(M) \tag{1.11}$$

commutes with the operators $\mathbf{H}_0$ and $\mathbf{S}^z$ but does not commute with the operator $\mathbf{H}_1$, and, thus, the total Hamiltonian H does not preserve the number of "a-fermions." At the same time, the operator $(-1)^{\mathbf{N}} = \exp\{\pm i\pi\mathbf{N}\}$, anticommuting with the creation and annihilation operators

$$\left[(-1)^{\mathbf{N}}, a_m^+\right]_+ = \left[(-1)^{\mathbf{N}}, a_m\right]_+ = 0, \tag{1.12}$$

commutes with any operator bilinear in a_m and a_m^+ and, in particular, with the Hamiltonian

$$\left[(-1)^{\mathbf{N}}, \mathbf{H}\right] = 0. \tag{1.13}$$

Periodic boundary conditions (1.6) for the spins lead to the following conditions for the fermions:

$$a_{M+1} = (-1)^{\mathbf{N}} a_1, \quad a_{M+1} = a_1^+ (-1)^{\mathbf{N}}. \tag{1.14}$$

Introducing the projectors $\mathbf{P}^{\pm}$,

$$\begin{gathered} \mathbf{P}^{\pm} = \frac{1}{2}(1 \pm (-1)^{\mathbf{N}}), \\ (\mathbf{P}^{\pm})^2 = \mathbf{P}^{\pm}, \quad \mathbf{P}^+ + \mathbf{P}^- = 1, \quad \mathbf{P}^+\mathbf{P}^- = \mathbf{P}^-\mathbf{P}^+ = 1, \\ \mathbf{P}^{\pm} a_m = a_m \mathbf{P}^{\mp}, \quad [\mathbf{H}, \mathbf{P}^{\pm}] = 0, \end{gathered} \tag{1.15}$$

we can rewrite the Hamiltonian in the following form [23]:

$$\mathbf{H} = \mathbf{H}^+\mathbf{P}^+ + \mathbf{H}^-\mathbf{P}^-. \tag{1.16}$$

The operators $\mathbf{H}^{\pm}$ can formally be rewritten in the same form

$$\mathbf{H}^{\pm} = -\frac{1}{2}\sum_{m=1}^{M}\left[(a_m^+ a_{m+1} + a_{m+1}^+ a_m) + \gamma(a_{m+1} a_m + a_m^+ a_{m+1}^+)\right] + h\sum_{m=1}^{M} a_m^+ a_m - \frac{hM}{2}, \tag{1.17}$$

and the only distinction between $\mathbf{H}^+$ and $\mathbf{H}^-$ is in the boundary conditions,

$$\begin{aligned} &a_{M+1} = -a_1, \quad a_{M+1}^+ = -a_1^+ \quad \text{for } \mathbf{H}^+ \\ &a_{M+1} = a_1, \quad\ \ a_{M+1}^+ = a_1^+ \quad\ \ \text{for } \mathbf{H}^-. \end{aligned} \tag{1.18}$$

Hence the Fourier transformations leading to the momenta representation are different for these Hamiltonians. We denote the sets of permitted quasimomenta by X^+ for $\mathbf{H}^+$ and by X^- for $\mathbf{H}^-$,

$$X^{\pm} = \{p : \exp\{ipM\} = \mp 1, \quad p \in (-\pi, \pi]\}, \tag{1.19}$$

or more explicitly,

$$X^+ = \left\{p_l = -\pi - \frac{\pi}{M} + \frac{2\pi}{M} l, \ \ l = 1, 2, \ldots, M\right\} \quad X^- = \left\{p_l = -\pi + \frac{2\pi}{M} l, \ \ l = 1, 2, \ldots, M\right\}. \tag{1.20}$$

The corresponding formulas for the Fourier transformation can be written in the following form:

$$a_m = \frac{\exp\{-i\pi/4\}}{\sqrt{M}} \sum_{p\in X^{\pm}} a_p \exp\{i(m-1)p\}, \quad a_m^+ = \frac{\exp\{i\pi/4\}}{\sqrt{M}} \sum_{p\in X^{\pm}} a_p^+ \exp\{-i(m-1)p\} \tag{1.21}$$

(the summation is taken over $p \in X^+$ for $\mathbf{H}^+$ and over $p \in X^-$ for $\mathbf{H}^-$) and

$$a_p = \frac{\exp\{i\pi/4\}}{\sqrt{M}} \sum_{m=1}^{M} a_m \exp\{-i(m-1)p\}, \quad a_p^+ = \frac{\exp\{-i\pi/4\}}{\sqrt{M}} \sum_{m=1}^{M} a_m^+ \exp\{i(m-1)p\}. \tag{1.22}$$

In the momenta representation, the Hamiltonians $\mathbf{H}^{\pm}$ have the following form:

$$\mathbf{H}^{\pm} = \sum_{p\in X^{\pm}} \left[\varepsilon(p)\, a_p^+ a_p + \frac{\Gamma(p)}{2}(a_p^+ a_{-p}^+ + a_{-p}\, a_p)\right] - \frac{Mh}{2}, \tag{1.23}$$

where

$$\varepsilon(p) = h - \cos p, \quad \Gamma(p) = \gamma \sin p. \tag{1.24}$$

The diagonalization of these Hamiltonians can be performed by the Bogolyubov transformations (different for $\mathbf{H}^+$ and $\mathbf{H}^-$) leading to new canonical fermion operators A_p and A_p^+:

$$A_p = \alpha(p)\, a_p - \beta(p)\, a_{-p}^+, \quad A_p^+ = \beta(p)\, a_p + \alpha(p)\, a_{-p}^+, \tag{1.25}$$

where

$$\alpha(p) = \cos\frac{\theta(p)}{2}, \quad \beta(p) = \sin\frac{\theta(p)}{2}, \tag{1.26}$$

and the angle θ is defined by the relations

$$\cos\theta(p) = \frac{\varepsilon(p)}{E(p)}, \quad \sin\theta(p) = -\frac{\Gamma(p)}{E(p)} \quad (p \neq 0, \pi, \quad \theta(p) = -\theta(-p), \tag{1.27}$$

$$E(p) = \sqrt{\varepsilon^2(p) + \Gamma^2(p)} \geq 0 \quad (p \neq 0.\pi). \tag{1.28}$$

The momenta $p = 0, \pi$ (appearing only in $\mathbf{H}^-$) need special consideration. Following [23], we put

$$A_0 = a_0, \quad A_0^+ = a_0^+, \quad A_\pi = a_\pi, \quad A_\pi^+ = a_\pi^+ \tag{1.29}$$

and

$$E(0) = h - 1 = +\varepsilon(0) \quad (E(0) < 0 \text{ for } h < 1), \quad E(\pi) = h + 1 = +\varepsilon(\pi) > 0. \tag{1.30}$$

The Hamiltonians $\mathbf{H}^{\pm}$ can be diagonalized as follows:

$$\mathbf{H}^{\pm} = \sum_{p\in X^{\pm}} E(p)\, A_p^+ A_p + E_0^{\pm}, \tag{1.31}$$

where the "vacuum energy" appears as

$$E_0^{\pm} = -\frac{1}{2} \sum_{p\in X^{\pm}} E(p) \tag{1.32}$$

(to calculate E_0^-, one should take into account definition (1.30)). It should be noted that, for $h < 1$, the value E_0^- is not the ground state energy E_g^-, and, in this case, we have $E_g^- = E_0^- + \varepsilon(0)$.

2. Coherent states for the XXO chain

In this section, we introduce some formulas for the matrix elements of operators between the coherent states of the isotropic XXO chain. These formulas will be used in the subsequent calculations. In this section, we simplify the notation and denote the corresponding Hamiltonians $\mathbf{H}^{\pm}_{\mathrm{XXO}}$ (see 1.23),

$$\mathbf{H}^{\pm}_{\mathrm{XXO}} = \sum_{p\in X^{\pm}} \varepsilon(p)\, a_p^+ a_p - \frac{hM}{2}, \tag{2.1}$$

by $\mathbf{H}^{\pm}$. They are already diagonal in terms of the operators a_p and a_p^+ (1.21). However, it should be noted that the Fock vacuum (the same for $\mathbf{H}^{\pm}$),

$$\begin{aligned} &a_m \mid 0\rangle = 0, \quad \langle 0 \mid a_m^+ = 0 \quad (m = 1, 2, \dots, M), \\ &a_p \mid 0\rangle = 0, \quad \langle 0 \mid a_p^+ = 0 \quad (p \in X^{\pm}), \qquad\qquad \langle 0 \mid 0\rangle = 1 \end{aligned}$$

is the ground state only for $h > 1$.

We introduce different coherent states (see Appendix A) for the Hamiltonians $\mathbf{H}^+$ and $\mathbf{H}^-$, namely,

$$\mid \phi, \pm\rangle = \exp\Big\{ \sum_{q\in X^{\pm}} a_q^+ \phi_q \Big\} \mid 0\rangle, \tag{2.3}$$

$$\langle \phi^*, \pm \mid = \langle 0 \mid \exp\Big\{ \sum_{q\in X^{\pm}} \phi_q a_q \Big\}. \tag{2.4}$$

The parameters φ_q and φ_q^* (Grassmann algebra elements) anticommute with each other and with the operators a and a^+. The main properties of the coherent states (2.3) and (2.4) are described in Appendix A. They are eigenstates for the operators a_p and a_p^+:

$$a_p \mid \phi, \pm\rangle = \phi_p \mid \phi, \pm\rangle \quad (p \in X^{\pm}), \tag{2.5}$$

$$\langle \phi^*, \pm \mid a_p^+ = \phi_p^* \langle \phi^*, \pm \mid \quad (p \in X^{\pm}). \tag{2.6}$$

The scalar product of coherent states of the same type is given by the usual formulas (A.4):

$$\langle \phi^*, + \mid \phi, + \rangle = \exp\Big\{ \sum_{p\in X^+} \phi_p^* \phi_p \Big\}, \tag{2.7}$$

$$\langle \phi^*, - \mid \phi, - \rangle = \exp\Big\{ \sum_{q\in X^-} \phi_q^* \phi_q \Big\}. \tag{2.8}$$

When it does not lead to confusion, we use the following notation for the sums on the right-hand sides of Eqs. (2.3), (2.4), (2.7), and (2.8): $a^+\varphi = \sum a_q^+ \varphi_q$, $\varphi^+\varphi = \sum \varphi_p^+ \varphi_p$, etc.

The scalar products of the coherent states of different type are given by the formulas

$$\begin{aligned} \langle \phi^*, + \mid \psi, - \rangle &= \exp\Big\{ \sum_{p\in X^+, q\in X^-} \phi_p^* L_{pq} \psi_q \Big\} = \exp\{\phi^* L \psi\} \langle \psi^*, - \mid \phi^*, + \rangle \\ &= \exp\Big\{ \sum_{p\in X^-, q\in X^+} \psi_p^* L_{pq} \phi_q \Big\} = \exp\{\psi^* L \phi\}, \end{aligned} \tag{2.9}$$

where the entries of the $M \times M$ matrix L have the form

$$L_{pq} = \frac{2}{M} \frac{1}{1 - \exp\{-i(p-q)\}} = \frac{i}{M} \left(\cot \frac{q-p}{2} - i \right). \tag{2.10}$$

This becomes clear if one takes into account that the Fock vacuum is the same for both types of states and rewrites the scalar products in the "coordinate representation" by formulas (1.21) and (1.22). For example,

$$\langle \phi^*, + \mid \psi, - \rangle = \langle 0 \mid \exp\Big\{ \sum_{m=1}^{M} \phi_m^* \, a_m \Big\} \exp\Big\{ \sum_{m=1}^{M} a_m^+ \, \psi_m \Big\} \mid 0 \rangle = \exp\Big\{ \sum_{m=1}^{M} \phi_m^* \, \psi_m \Big\}, \tag{2.11}$$

where

$$\phi_m^* = \frac{\exp(i\pi/4)}{\sqrt{M}} \sum_{p \in X^+} \phi_p^* \exp\{-i(m-1)p\} \psi_m^* = \frac{\exp(-i\pi/4)}{\sqrt{M}} \sum_{q \in X^-} \psi_q \exp\{i(m-1)1\}. \tag{2.12}$$

Now we consider the matrix elements of the operator $\exp\{\sigma_{m+1}^+ \mathbf{Q}(m)\}$, where (see (1.9)) $\mathbf{Q}(m)$ is the operator of the number of particles at the first m sites of the lattice:

$$\mathbf{Q}(m) = \sum_{l=1}^{m} a_l^+ \, a_l = \sum_{p_1, p_2} a_{p_1}^+ \, Q_{p_1 p_2}(m) \, a_{p_2} \equiv a^+ \, Q(m) \, a \tag{2.13}$$

(the quasimomenta p_1 and p_2 correspond to the states of the same type $p_1, \, p_2 \in X^+$ or $p_1, \, p_2 \in X^-$). Using the properties of the matrix $Q(m)$ (obvious in the coordinate representation),

$$Q^2(m) = Q(m), \quad \exp\{\alpha Q(m)\} = I + (e^\alpha - 1) \, Q(m), \tag{2.14}$$

we arrive at the following expressions for matrix elements between two states of the same type:

$$\langle \phi^*, \pm \mid \exp\{\alpha \mathbf{Q}(m)\} \mid \phi, \pm \rangle = \exp\big\{ \phi^* [I + (e^\alpha - 1) \, Q(m)] \phi \big\}$$

$$= \exp\Big\{ \sum_{p \in X^\pm, q \in X^\pm} \phi_p^* [\delta_{pq} + (e^\alpha - 1) Q_{pq}(m)] \phi_q \Big\}, \tag{2.15}$$

and the entries of the $M \times M$ matrix $Q(m)$ are given by the formulas

$$Q_{pq}(m) = \exp\{-i(m-1)p/2\} Q_{pq}^{(0)}(m) \exp\{i(m-1)q/2\}, \tag{2.16}$$

$$Q_{pq}^{(0)}(m) = \frac{1}{M} \frac{\sin \frac{m(p-q)}{2}}{\sin \frac{(p-q)}{2}} \tag{2.17}$$

(for the diagonal matrix elements, we use the l'Hôpital rule, $Q_{pp}(m) = Q_{pp}^{(0)}(m) = m/M$).

Similarly to (2.9), we obtain the following formulas for the states of different types:

$$\langle \phi^*, \pm \mid \exp\{\alpha \mathbf{Q}(m)\} \mid \phi, \mp \rangle = \exp\big\{ \phi^* [L + (e^\alpha - 1) Q(m)] \phi \big\}$$

$$= \exp\Big\{ \sum_{p \in X^\pm, q \in X^\mp} \phi_p^* [L_{pq} + (e^\alpha - 1) Q_{pq}(m)] \phi_q \Big\}, \tag{2.18}$$

where L_{pq} and $Q_{pq}(m)$ are given by formulas (2.10) and (2.16) (note that now if $p \in X^+$, then $q \in X^-$ and vice versa). In particular, for the operator $\exp\{i\pi \mathbf{Q}(m)\}$ in the Jordan–Wigner transformation, we have

$$\langle \phi^*, \pm \mid \exp\{i\pi \mathbf{Q}(m)\} \mid \phi, \mp \rangle = \exp\big\{ \phi^* \, L(m) \, \phi \big\} \exp\Big\{ \sum_{p \in X^\pm, q \in X^\mp} \phi_p^* \, L_{pq}(m) \, \phi_q \Big\}, \tag{2.19}$$

where

$$L_{pq}(m) = \exp\{-imp\} \, L_{pq} \, \exp\{imq\}. \tag{2.20}$$

Now we consider the matrix elements ("form-factors") of the local spins. Since

$$P^{\pm}\,\sigma_m^{\alpha} = \sigma_m^{\alpha}\,P^{\mp} \quad (\alpha = x, y, \qquad \alpha = \pm), \tag{2.21}$$

we need only the matrix elements of the local operators $\sigma_m^{\pm}$ between the states of different type. A direct calculation using (1.7), (1.22), and (2.18) yields the relation

$$\langle\, \phi^*, \pm \mid \sigma_m^{+} \mid \psi, \mp \,\rangle = \psi_m(\mp)\exp\{\phi^*\, L(m-1)\psi\} = \psi_m(\mp)\exp\Big\{\sum_{p\in X^{\pm}, q\in X^{\mp}} \phi_p^*\, L_{pq}(m-1)\,\psi_q\Big\},$$

$$\langle\, \phi^*, \pm \mid \sigma_m^{-} \mid \psi, \mp \,\rangle = \phi_m^*(\pm)\exp\{\phi^*\, L(m-1)\psi\}, \tag{2.22}$$

where we used the notation

$$\psi_m(\pm) = \frac{\exp\{-i\pi/4\}}{\sqrt{M}} \sum_{p\in X^{\pm}} \psi_p \exp\{i(m-1)p\}\phi_m^*(\pm) = \frac{\exp\{i\pi/4\}}{\sqrt{M}} \sum_{p\in X^{\pm}} \phi_p^* \exp\{-i(m-1)p\}. \tag{2.23}$$

It should be noted that the matrices L (2.10), $L(m)$ (2.20), and $Q(m)$ (2.16) are related by the following simple formula, which can be proved by direct calculation:

$$\sum_q L_{p_1 q}(m)\, L_{q p_2} = \delta_{p_1 p_2} - 2Q_{p_1 p_2}(m) \tag{2.24}$$

$$(p_1, p_2 \in X^{+}, \quad q \in X^{-} \text{ or } p_1, p_2 \in X^{-},\ q \in X^{+}).$$

Using these formulas, one can reproduce the time-dependent temperature correlation functions for the isotropic XXO chain obtained in [4, 5, 26]. Since this method is very elegant and simple, we present it in Appendix B. In the next sections, we shall consider the anisotropic XY chain.

3. Coherent states for the anisotropic XY chain

In this section, we derive formulas for the matrix elements of the operators between the coherent states of the anisotropic XY chain. First we consider two sets of canonical fermion operators a_p, a_p^+ and A_p, A_p^+ related by the Bogolyubov transformation (1.25). For each of these sets, we introduce the coherent states ("old" and "new" ones) and calculate the scalar products of the states from the different sets. This gives a universal method of calculating the matrix elements of operators.

We begin with the relations between two vacuum states, $|0\rangle$ and $|0\rangle\rangle$, for the old and new sets:

$$a_p \mid 0\rangle = 0, \quad \langle 0 \mid a_p^+ = 0 \ \text{ for all } \ p, \quad \langle 0 \mid 0\rangle = 1, \tag{3.1}$$

$$A_p \mid 0\rangle\rangle = 0, \quad \langle\langle 0 \mid A_p^+ = 0, \quad \langle\langle 0 \mid 0\rangle\rangle = 1. \tag{3.2}$$

These states are related as follows:

$$|\, 0\rangle\rangle = N^{-1/2}\,\Omega^+ \mid 0\rangle, \quad \langle\langle 0 \mid = N^{-1/2}\langle 0 \mid \Omega, \tag{3.3}$$

where the operators Ω^+ and Ω are given by the formulas

$$\Omega^+ = \exp\Big\{\frac{1}{2}\sum_p \tau(p)\, a_p^+\, a_{-p}^+\Big\}, \quad \Omega = \exp\Big\{\frac{1}{2}\sum_p \tau(p)\, a_{-p}\, a_p\Big\}, \tag{3.4}$$

where

$$\tau(p) \equiv \tan\frac{\theta(p)}{2}, \tag{3.5}$$

and the normalization coefficient N can be represented as a determinant of a diagonal $M \times M$ matrix,

$$N = \langle 0 \mid \Omega^+ \Omega \mid 0 \rangle = \det (I + T), \tag{3.6}$$

where I is the identity matrix and

$$T = \operatorname{diag} (i\tau(p)). \tag{3.7}$$

For the states (3.3), properties (3.2) can easily be checked by the commutation relations

$$\left[a_p, \Omega^+\right] = \tau(p)\Omega^+ a^+_{-p} \quad \text{and} \quad \left[\Omega^+, a^+_p\right] = \tau(p)\, a_{-p}\, \Omega.$$

To calculate the normalization coefficient, we use formula (A.6) and obtain

$$N = \int d\xi\, d\xi^* \exp\left\{ -\xi^*\xi + \frac{1}{2}\tau(p)(\xi^*_p\, \xi^*_{-p} + \xi_{-p}\xi_p)\right\}.$$

We use the change of variables

$$\omega_p = \frac{1}{\sqrt{2}}(\xi_p + i\xi^*_{-p}), \quad \omega^*_p = \frac{1}{\sqrt{2}}(\xi^*_p + i\xi_{-p}). \tag{3.8}$$

The Jacobian of this transformation is equal to 1, $d\xi d\xi^* = d\omega d\omega^*$. It is easy to check that

$$\xi^*\xi = \omega^*\omega, \quad \frac{1}{2}\sum_p \tau(p)(\xi^*_p\, \xi^*_{-p} + \xi_{-p}\, \xi_p) = -\omega^*\, T\omega, \tag{3.9}$$

where the matrix T is defined in (3.7). Hence, $N = \int d\omega d\omega^* \exp\{-\omega^*(I+T)\omega\} = \det (I+T)$, which proves relation (3.6).

Now we introduce the coherent states for the new set of operators (cf. (A.2)):

$$\mid X \rangle\rangle = \exp\Big\{\sum_p A^+_p\, X_p\Big\} \mid 0 \rangle\rangle \quad \langle\langle X \mid = \langle\langle 0 \mid \exp\Big\{\sum_p X^*_p\, A_p\Big\}. \tag{3.10}$$

The Grassmann algebra elements X_p and X^*_p have the same properties as the old parameters ξ_p and ξ^*_p. Formulas (A.2)–(A.10) are also valid for the new states. A direct calculation gives the following representations for the scalar products of the old and new states:

$$\begin{aligned}
\langle \xi^* \mid X \rangle\rangle &= N^{-1/2} \exp\left\{\frac{1}{2}\sum_p \tau(p)(\xi^*_p\, \xi^*_{-p} - X_{-p}\, X_p) + \sum_p \alpha^{-1}(p)\, \xi^*_p\, X_p\right\}, \\
\langle X^* \mid \xi \rangle &= N^{-1/2} \exp\left\{\frac{1}{2}\sum_p \tau(p)(\xi_{-p}\, \xi_p - X^*_p\, X_{-p}) + \sum_p \alpha^{-1}(p)\, X^*_p\, \xi_p\right\}.
\end{aligned} \tag{3.11}$$

Now we consider the XY model. Here it is necessary to introduce different sets of operators a_p, a^+_p and A_p, A^+_p corresponding to the sets of momenta X_+ (for $\mathbf{H}^+$) and X_- (for $\mathbf{H}^-$) (see (1.19) and (1.20)). Thus, the new vacuums $|\Omega\rangle\rangle$ will also be different for $\mathbf{H}^+$ and $\mathbf{H}^-$ (there is a difference with the old vacuum $|0\rangle\rangle$). Usually, we shall not indicate this difference in our notation, but this must always be kept in mind.

Using formulas (3.11), we can calculate the matrix elements of the operators $\exp\{-\mathbf{H}^\pm\}$, which are diagonal in the new representations (different for $\mathbf{H}^+$ and $\mathbf{H}^-$!), between old coherent states (also different for $\mathbf{H}^+$ and $\mathbf{H}^-$),

$$\langle \xi^* \mid e^{-\beta \mathbf{H}^\pm} \mid \xi \rangle = N^{-1}\, e^{-\beta E^\pm_0} \det (I - J_\beta\, T) e^{\omega^*\, D\omega}, \tag{3.12}$$

where the variables ω_p and ω^*_p ($p \in X^+$ for $\mathbf{H}^+$ and $p \in X^-$ for $\mathbf{H}^-$) are defined by formula (3.8), and the diagonal $M \times M$ matrices J_β and D have the form

$$J_\beta = \operatorname{diag} (j_\beta(p)), \quad j_\beta(p) = \exp\{-\beta\, E(p)\}, \tag{3.13}$$

$$D = \operatorname{diag}(d(p)), \quad d(p) = \frac{j_\beta(p) - i\tau(p)}{1 - i\tau(p) j_\beta(p)} \tag{3.14}$$

(the matrix T is defined in (3.7)).

For the proof, we use the completeness of the new states (A.6),

$$\langle \xi^* \mid e^{-\beta \mathbf{H}^\pm} Y \mid \xi \rangle = \int dX\, dX^*\, dY\, dY^* \langle \xi^* \mid X \rangle\rangle\langle\langle Y^* \mid e^{-\beta \mathbf{H}^\pm} \mid\rangle\rangle\langle\langle X^* \mid \xi \rangle \exp\{-Y^* X - X^* Y\},$$

and note that the operators $\mathbf{H}^\pm$ are diagonal in the corresponding new representations (1.31),

$$\langle\langle Y^* \mid e^{-\beta \mathbf{H}^\pm} Y \mid\rangle\rangle = e^{-\beta E_0^\pm} \exp\Big\{ \sum_{p \in X^\pm} e^{-\beta E(p)} Y_p^* Y_p \Big\}. \tag{3.15}$$

After the integration over Y, Y^*, we obtain

$$\langle \xi^* \mid e^{-\beta \mathbf{H}^\pm} \mid \xi \rangle = \int dX\, dX^* \langle \xi^* \mid X \rangle\rangle\langle\langle X^* \mid \xi \rangle e^{-\beta E_0^\pm} \exp\Big\{ \sum_{p \in X^\pm} e^{-\beta E(p)} X_p^* X_p \Big\}.$$

The integral over X, X^* can be calculated by a change of variables similar to (3.8). Finally, we obtain formula (3.12).

In the next sections, the results obtained will be used for calculating the equal-time correlators.

4. The simplest correlator for the XY chain

In this section, the partition function Z and the generating functional of the third components of local spins will be calculated for the anisotropic chain. We begin with the partition function. The initial representation is the same as in the isotropic case (B.1):

$$Z = \frac{1}{2}(Z_F^+ + Z_F^- + Z_B^+ - Z_B^-), \tag{4.1}$$

where

$$Z_F^\pm = \operatorname{Sp} \exp\{-\beta \mathbf{H}^\pm\}, \quad Z_B^\pm = \operatorname{Sp} \left(\exp\{-\beta \mathbf{H}^\pm\}(-1)^{\mathbf{N}}\right). \tag{4.2}$$

We obtain the following representations for the contributions (see, e.g., [25]; the $M \times M$ matrix J_β is defined in (3.13)):

$$Z_F^\pm = e^{-\beta E_0^\pm} \det(I + J_\beta) = \prod_{p \in X^\pm} \left(2\cosh \frac{\beta E(p)}{2}\right), \tag{4.3}$$

$$Z_B^\pm = e^{-\beta E_0^\pm} \det(I + J_\beta) = \prod_{p \in X^\pm} \left(2\sinh \frac{\beta E(p)}{2}\right). \tag{4.4}$$

To obtain the fermion contributions, we use representation (A.5) for the trace of operators,

$$Z_F^\pm = \int dY\, dY^* \langle\langle Y^* \mid \beta^{-\beta \mathbf{H}^\pm} \mid Y \rangle\rangle \exp\{Y^* Y\}$$

and representation (3.15) for the matrix element occuring here; thereafter it remains to calculate the Gauss integral and arrive at Eq. (4.3). The "boson" contributions can be calculated similarly.

It should be noted that it is sometimes convenient to represent the answer differently, by using the old coherent states, for example,

$$Z_F^\pm = \int d\xi\, d\xi^* \langle \xi^* \mid \beta^{-\beta \mathbf{H}^\pm} \mid \xi \rangle \exp\{\xi^* \xi\},$$

and formula (3.12) for the corresponding matrix element. We obtain

$$Z_F^{\pm} = N^{-1}\, e^{-\beta E_0^{\pm}} \det(I - J_\beta T) \det(I + D), \tag{4.5}$$

$$Z_B^{\pm} = N^{-1}\, e^{-\beta E_0^{\pm}} \det(I - J_\beta T) \det(I - D). \tag{4.6}$$

Now consider the simplest equal-time temperature correlator

$$G(m) = \frac{1}{Z}\mathrm{Sp}\left(e^{\alpha \mathbf{Q}(m)}\, e^{-\beta \mathbf{H}}\right). \tag{4.7}$$

As for the isotropic chain ((B.8) and (B.9)), this correlator can be represented as a sum of four contributions:

$$G(m) = \frac{1}{2Z}\left(Z_F^{+} G_F^{+} + Z_F^{-} G_F^{-} + Z_B^{+} G_B^{+} - Z_B^{-} G_B^{-}\right), \tag{4.8}$$

where

$$Z_F^{\pm} G_F^{\pm} = \mathrm{Sp}\left(e^{\alpha \mathbf{Q}(m)}\, e^{\beta \mathbf{H}^{\pm}}\right), \quad Z_B^{\pm} G_B^{\pm} = \mathrm{Sp}\left(e^{\alpha \mathbf{Q}(m)}\, e^{\beta \mathbf{H}^{\pm}} (-1)^{\mathbf{N}}\right). \tag{4.9}$$

We have the following representations of the contributions $G_{F,B}^{\pm}$ in the form of determinants of $M \times M$ matrices:

$$G_F^{\pm} = \det\left(I + (e^{\alpha} - 1) Q^{(0)}(m)\, \Omega_F\right), \tag{4.10}$$

$$G_B^{\pm} = \det\left(I - (e^{\alpha} - 1) Q^{(0)}(m)\, \Omega_B\right). \tag{4.11}$$

Here the matrix elements of $Q^{(0)}(m)$ are given by (2.16), $p, q \in X^{+}$ for $G_{F,B}^{+}$, and $p, q \in X^{-}$ for $G_{F,B}^{-}$. The diagonal $M \times M$ matrices Ω_F and Ω_B are given by the following formulas ($p \in X^{\pm}$ for $G_{F,B}^{\pm}$):

$$\Omega_F = D(I + D)^{-1} = \mathrm{diag}\,(\omega_F(p)), \quad \omega_F(p) = \frac{1}{2}\left(1 - e^{i\theta(p)} \tanh \frac{\beta E(p)}{2}\right), \tag{4.12}$$

$$\Omega_B = D(I - D)^{-1} = \mathrm{diag}\,(\omega_B(p)), \quad \omega_B(p) = \frac{1}{2}\left(e^{i\theta(p)} \coth \frac{\beta E(p)}{2} - 1\right), \tag{4.13}$$

(the matrix D is defined in (3.14)).

Now we show how to calculate the contribution G_F^{+}. We have

$$Z_F^{+} G_F^{+} = \int d\xi\, d\xi^*\, d\eta\, d\eta^* \langle \eta^* \mid e^{\alpha \mathbf{Q}(m)} \mid \eta \rangle \langle \xi^* \mid e^{-\beta \mathbf{H}^{+}} \mid \xi \rangle e^{\eta^* \xi - \xi^* \eta}. \tag{4.14}$$

The matrix elements on the right-hand side are given by (2.15) and (3.12). We make the change of variables $\eta^* \to -\eta^*$. Since M is even, the integration measure is invariant. Next we change the variables as in (3.8):

$$\begin{aligned} \omega_p &= \frac{1}{\sqrt{2}}(\xi_p + i\xi_{-p}^*), \quad \omega_p^* = \frac{1}{\sqrt{2}}(\xi_p^* + i\xi_{-p}), \\ \rho_p &= \frac{1}{\sqrt{2}}(\eta_p + i\eta_{-p}^*), \quad \rho_p^* = \frac{1}{\sqrt{2}}(\eta_p + i\eta_{-p}^*). \end{aligned} \tag{4.15}$$

The measure is invariant under this change of variables, and

$$\eta^* \xi + \xi^* \eta = \rho^* \omega + \omega^* \rho, \quad \eta^* \big(I + (e^{\alpha} - 1) Q(m)\big) \eta = \rho^* \big(I + (e^{\alpha} - 1) Q(m)\big) \rho. \tag{4.16}$$

The integration over ω, ω^* of the factors depending on these variables in (4.14) gives $\det D \exp\{-\rho^* D^{-1} \rho\}$. Taking into account representation (4.5) for the partition function and (4.12), we can calculate the remaining Gauss integral over ρ, ρ^*:

$$Z_F^{+} G_F^{+} = Z_F^{+} \det\left(I + (e^{\alpha} - 1)\, Q(m)\, \Omega_F\right). \tag{4.17}$$

Note that the matrix $Q(m)$ (2.16) differs from the matrix $Q^{(0)}(m)$ only by an obvious similarity transformation with a diagonal matrix. Thus, representation (4.10) for G_F^+ is proved. The contribution of G_F^- can be calculated in the same way, but one should use the momenta from the set X^-. To calculate the boson contributions, one should use the following property of the operator $(-1)^{\mathbf{N}}$:

$$(-1)^{\mathbf{N}} \mid \xi \rangle = \mid -\xi \rangle \tag{4.18}$$

in (4.14), which obviously changes the subsequent calculations and leads to (4.11).

One additional important remark concerning formulas (4.10) and (4.11) is necessary. Since the sets X^+ and X^- are symmetric with respect to the change of momenta $p_i \to -p_i$ (an exception is provided by the momenta 0 and π from the set X^-, but the result is also valid in this case), one can rewrite the answer in the form (the sign "+" on the right-hand side corresponds to G_F, and the sign "−" corresponds to G_B)

$$\begin{aligned} G_{F,B}^{\pm} &= \det\left(I \pm (e^{\alpha}-1)\, Q(m)\, \Omega_{F,B}\right) = \det\left(I \pm (e^{\alpha}-1)\, \Omega_{F,B}^{1/2}\, Q(m)\, \Omega_{F,B}^{1/2}\right) \\ &= \det\left(I \pm (e^{\alpha}-1)\, \overline{\Omega}_{F,B}^{1/2}\, Q(m)\, \overline{\Omega}_{F,B}^{1/2}\right) = \det\left(I \pm (e^{\alpha}-1)\, Q(m)\, \overline{\Omega}_{F,B}\right). \end{aligned} \tag{4.19}$$

Here the bar means complex conjugation, and $\omega_{F,B}(-p) = \overline{\omega}_{F,B}(p)$.

In conclusion, we discuss some limit cases.

In the zero-temperature limit ($\beta \to \infty$), the "odd" contributions to the correlator are cancelled, and we obtain the following representation for $G(m)$:

$$G(m) = \det\left(I + (e^{\alpha}-1)\, Q^{(0)}(m)\, \Omega_0\right) \quad (T=0), \tag{4.20}$$

where

$$\Omega_0 = \operatorname{diag}\left(\omega_0(p)\right), \quad \omega_0(p) = \frac{1}{2}\left(1 - e^{i\theta(p)}\right). \tag{4.21}$$

This coincides with the result obtained in [27]. On the other hand, for the isotropic case ($\gamma = 0$), taking into account that the angle $\theta(p)$ in the Bogolyubov transformation is

$$\theta(p) = -\pi \operatorname{sgn} p, \quad |p| < k_F, \quad \theta(p) = 0, \quad |p| > k_F \quad (\gamma = 0), \tag{4.22}$$

we obtain the following formulas for the weights:

$$\begin{aligned} \omega_F(p) &= \frac{1}{2}\left(1 - \tanh\frac{\beta\varepsilon(p)}{2}\right), \\ \omega_B(p) &= \frac{1}{2}\left(\coth\frac{\beta\varepsilon(p)}{2} - 1\right) \quad (\gamma = 0). \end{aligned} \tag{4.23}$$

Here $k_F = \arccos h$ is the Fermi momentum, and $\varepsilon(p)$ is the dispersion of the XXO chain (see (1.24)). Thus, the answers for the isotropic case [4, 26] are reproduced.

5. Equal-time correlators of the local spins

Here we consider the equal-time correlation functions of the local spin operators ($\sigma_{m+1}^- \equiv 1/T$):

$$G^{(ab)}(m) = \langle\, \sigma_{m+1}^a\, \sigma_1^b \,\rangle_T = \frac{1}{Z} \operatorname{Sp}\left(\sigma_{m+1}^a\, \sigma_1^b\, e^{-\beta \mathbf{H}}\right), \quad a, b = +, -. \tag{5.1}$$

On a finite lattice of length M, these correlators can be represented (as in (B.8)) as a sum of four contributions:

$$G^{(ab)} = \frac{1}{2Z}\left(Z_F^+\, G_F^{(ab),+} + Z_F^-\, G_F^{(ab),-} + Z_B^+\, G^{(ab),+} - Z_B^-\, G^{(ab),-}\right), \tag{5.2}$$

where

$$Z_F^{\pm} G_F^{(ab),\pm} = \mathrm{Sp}\left(\sigma_{m+1}^a \sigma_1^b e^{-\beta \mathbf{H}^{\pm}}\right); \quad Z_B^{\pm} G_B^{(ab),\pm} = \mathrm{Sp}\left(\sigma_{m+1}^a \sigma_1^b e^{-\beta \mathbf{H}^{\pm}}, (-1)^{\mathbf{N}}\right) \tag{5.3}$$

(the partition functions $Z_{F,B}^{\pm}$ are given by (4.3) and (4.4)). For the contributions, we obtain the following representations in the form of determinants of $M \times M$ matrices:

$$G_F^{(-+),\pm} = G_F^{(+-),\pm} = \frac{\partial}{\partial \alpha} \det\left(I + U\Omega_F + \alpha C \Omega_F\right)\big|_{\alpha=0} \ (m > 0), \tag{5.4}$$

$$G_F^{(-+),\pm} = \frac{1}{M} \mathrm{tr}\, \Omega_F = \frac{1}{M} \sum_p \omega_F(p) \ (m = 0), \quad G_F^{(+-),\pm} = 1 - \frac{1}{M} \mathrm{tr}\, \Omega_F \ (m = 0), \tag{5.5}$$

$$G_F^{(++),\pm} = G_F^{(--),\pm} = \frac{\partial}{\partial \alpha} \det\left(I + U\Omega_F - i\alpha S \Omega_F\right)\big|_{\alpha=0} \ (m \geq 0), \tag{5.6}$$

$$G_B^{(-+),\pm} = G_B^{(+-),\pm} = \frac{\partial}{\partial \alpha} \det\left(I - U\Omega_B - \alpha C \Omega_B\right)\big|_{\alpha=0} \ (m \geq 0), \tag{5.7}$$

$$G_B^{(-+),\pm} = -\frac{1}{M} \mathrm{tr}\, \Omega_B, \ G_B^{(+-),\pm} = 1 + \frac{1}{M} \mathrm{tr}\, \Omega_B \ (m = 0), \tag{5.8}$$

$$G_B^{(++),\pm} = G_B^{(--),\pm} = \frac{\partial}{\partial \alpha} \det\left(I - U\Omega_B - i\alpha S \Omega_B\right) \quad (m \geq 0). \tag{5.9}$$

Diagonal $M \times M$ matrices of weights Ω_F and Ω_B are defined in (4.12) and (4.13). The matrix elements of $M \times M$ matrices U, C, and S are given by the following formulas:

$$U_{p_1 p_2} = -\exp\left\{\frac{i}{2}(p_1 - p_2)\right\} Q^{(0)}_{p_1 p_2}(m) \tag{5.10}$$

(see the definition of $Q^{(0)}(m)$ in (2.17)),

$$C_{p_1 p_2} = \frac{1}{M} \cos \frac{m}{2}(p_1 + p_2), \quad \widetilde{C}_{p_1 p_2} = C_{p_1 p_2}(1 - 2\delta_{m,0}) \tag{5.11}$$

$$S_{p_1 p_2} = \frac{1}{M} \sin \frac{m}{2}(p_1 + p_2). \tag{5.12}$$

It should be noted that the momenta numerating the matrix elements are $p_1, p_2 \in X^+$ for the contributions of $G_{F,B}^{(ab),+}$ and $p_1, p_2 \in X^-$ for the contributions of $G_{F,B}^{(ab),-}$; the same is true for the weight matrices.

To calculate the functions $G_{F,B}^{(ab),\pm}$, one should calculate the Gauss integrals over the Grassmannian variables. For example, using (A.5) and (A.6), we obtain

$$Z_F^{+} G^{(ab),+} = \int d\xi\, d\xi^*\, d\eta\, d\eta^* \langle \eta^* \mid \sigma_{m+1}^a \sigma_1^b \mid \eta \rangle \langle \xi^* \mid e^{-\beta \mathbf{H}^{\pm}} \mid \xi \rangle e^{\eta^* \xi - \xi^* \eta}. \tag{5.13}$$

Now we calculate the matrix element

$$F_{ab} = \langle \eta^* \mid \sigma_{m+1}^a \sigma_1^b \mid \eta \rangle = \int d\zeta\, d\zeta^* \langle \eta^* \mid \sigma_{m+1}^a \mid \zeta \rangle \langle \zeta^* \mid \sigma_1^b \mid \eta \rangle e^{-\zeta^* \zeta}.$$

We use formulas (2.22) for the matrix elements of spin operators and change the variables:

$$\widetilde{\eta_p}^* = e^{-\frac{im}{2}p} \eta_p^*, \quad \widetilde{\eta}_p = e^{\frac{im}{2}p} \eta_p, \quad (p \in X^+), \quad \widetilde{\zeta_q}^* = e^{-\frac{im}{2}q} \zeta_q^*, \quad \widetilde{\zeta}_q = e^{\frac{im}{2}p} \zeta_q, \quad (q \in X^-). \tag{5.14}$$

We obtain the following representation of the matrix element in the new variables (we omit the tildes over the new variables (5.14)):

$$F_{ab} = \frac{\partial}{\partial \alpha} \int d\zeta\, d\zeta^* \exp\{\omega + \alpha f_{ab}\}\big|_{\alpha=0}, \tag{5.15}$$

where

$$\omega = \eta^* PL\overline{P}\zeta + \zeta^*\overline{P}LP\eta - \zeta^*\zeta, \tag{5.16}$$

$$f_{-+} = \eta^* R_+ \eta,\ (R_+)_{p_1p_2} = \frac{1}{M} e^{-\frac{im}{2}(p_1+p_2)}, \tag{5.17}$$

$$f_{+-} = -\zeta^* \overline{R}_+ \zeta,\ (\overline{R}_+)_{q_1q_2} = \frac{1}{M} e^{\frac{im}{2}(q_1+q_2)}, \tag{5.18}$$

$$f_{++} = i\eta R_- \zeta,\ (R_-)_{pq} = \frac{1}{M} e^{-\frac{im}{2}(p-q)}, \tag{5.19}$$

$$f_{--} = -\zeta^* \overline{R}_- \eta^*,\ (\overline{R}_-)_{qp} = \frac{1}{M} e^{\frac{im}{2}(q-p)} = (R_-)_{pq} \tag{5.20}$$

(the bar means complex conjugation). The diagonal matrix P has the form

$$P = \operatorname{diag}\left(e^{-\frac{im}{2}p}\right). \tag{5.21}$$

We use the following identities:

$$\sum_q L_{pq}\, e^{imq} = (2\delta_{m,0} - 1)\, e^{ipm},\ \ \sum_q e^{imq}\, L_{qp} = e^{ipm}\ (p \in X^+,\ q \in X^-,\ m = 0, 1, \ldots, M-1), \tag{5.22}$$

$$(PL\overline{P}\,\overline{R}_+\overline{P}LP)_{p_1p_2} = -(\overline{R}_+)_{p_1p_2}\,(1 - 2\delta_{m,0}), \tag{5.23}$$

$$(R_-\overline{P}LP)_{p_1p_2} = (R_-)_{p_1p_2}, \tag{5.24}$$

$$(PL\overline{P}\,\overline{R}_-)_{p_1p_2} = -(1 - 2\delta_{m,0})(\overline{R}_-)_{p_1p_2}. \tag{5.25}$$

It should be noted that, on the left-hand sides of (5.23) and (5.24), we have matrices with the entries

$$(\overline{R}_+)_{q_1q_2}\ (q_{1,2} \in X^-) \text{ and } (R_-)_{pq}\ (p \in X^+,\ q \in X^-),$$

and, on the right-hand side, we have $(\overline{R}_+)_{p_1p_2}$, $(R_-)_{p_1p_2}(p_{1,2} \in X^+)$. As a result, the matrix elements F_{ab} (5.15) are represented as follows:

$$\begin{aligned}
F_{-+} &= \frac{\partial}{\partial\alpha} \exp\{\eta^*(I+U)\eta + \alpha\eta^* R_+\eta\}\,|_{\alpha=0}, \\
F_{+-} &= \left(\delta_{m,0} + \frac{\partial}{\partial\alpha}\right) \exp\left\{\eta^*(I+U)\eta + \alpha\eta^* \tilde{\overline{R}}_+\eta\right\}|_{\alpha=0}, \\
F_{++} &= \frac{\partial}{\partial\alpha} \exp\{\eta^*(I+U)\eta + \alpha\eta S_-\eta\}\,|_{\alpha=0}, \\
F_{--} &= \frac{\partial}{\partial\alpha} \exp\{\eta^*(I+U)\eta - \alpha\eta^* S_-\eta^*\}\,|_{\alpha=0}.
\end{aligned} \tag{5.26}$$

Here the entries of the $M \times M$ matrix U are defined as $U_{p_1p_2} = (PL\overline{P}^2LP)_{p_1p_2} - \delta_{p_1p_2}$, and this representation leads to (5.10) if one takes into account (2.24); the matrices R_+ and $\overline{R}_+$ are given by (5.17), (5.18), and (5.12); the matrix $\tilde{\overline{R}}_+$ is defined by the relation

$$(\tilde{\overline{R}}_+)_{p_1p_2} = (\overline{R}_+)_{p_1p_2}(1 - 2\delta_{m,0}), \tag{5.27}$$

$$(S_-)_{p_1p_2} = \frac{1}{M} \sin\frac{m}{2}(p_1 - p_2). \tag{5.28}$$

Now we substitute expressions (5.12) in (5.3) and use (3.12). Then, for the contribution of $G_F^{(-+),+}$, we obtain

$$Z_F^+ G_F^{(-+),+}$$
$$= N^{-1} e^{-\beta E_0^+} \det(I - J_\beta T) \frac{\partial}{\partial \alpha} \int d\xi\, d\xi^*\, d\eta\, d\eta^* \{-\eta^*(I + U + \alpha R_+)\eta + \omega^* D\omega - \eta^*\xi - \xi^*\eta\} \quad (5.29)$$

(in addition to (3.8), we have changed the variables as in (5.14) and used the change $\eta^* \to -\eta^*$). Finally, we change the variables as in (4.15), $(\eta, \eta^*) \to (\rho, \rho^*)$, $(\xi, \xi^*) \to (\omega, \omega^*)$. Using (4.16) and the equations

$$\eta^*(I + U + \alpha R_+)\eta = \rho^*(I + U + \alpha C)\rho - \frac{\alpha}{2}(\rho^* S_- \rho^* - \rho S_- \rho),$$

we integrate over ω, ω^* and then over ρ, ρ^* in (5.29). As a result, we obtain the following representation in the form of the determinant of a $2M \times 2M$ matrix:

$$Z_F^+ G_F^{(-+),+} = N^{-1} e^{-\beta E_0^+} \det(I - J_\beta T) \det D \frac{\partial}{\partial \alpha} \det{}^{1/2} \begin{pmatrix} \alpha S_- & \vdots & -B \\ \cdots & \cdots & \cdots \\ B^T & \vdots & -\alpha S_- \end{pmatrix},$$

where $B = I + D^{-1} + U + \sigma_{m+1}^+ C$. Using the generalized Gauss algorithm, one can easily check that

$$\det \begin{pmatrix} \alpha S_- & \vdots & -B \\ \cdots & \cdots & \cdots \\ B^T & \vdots & -\alpha S_- \end{pmatrix} = \det{}^2 B + O(\alpha^2)$$

(this is also valid for $m = 0$). As a result, we obtain representation (5.4) for $G_F^{(-+),+}$. Similarly, one can calculate the other contributions.

It should be noted that expressions (5.4)–(5.9) for equal-time temperature correlators of the anisotropic XY chain differ from the corresponding expressions for the isotropic XXO chain only by the weights. For example, for the "fermion" weight, the corresponding modification, as before, has the form (5.30).

6. The thermodynamic limit

Now we consider the correlators for the anisotropic XY chain in the thermodynamic limit (the length of the chain $M \to \infty$, the magnetic field h is fixed). In this limit, the partition functions $Z_{F,B}^{\pm}$ diverge, and, using (4.3) and (4.4), one can obtain the estimate

$$Z_B / Z_F < C^M, \quad C = \tanh \frac{\beta E_{\max}}{2} < 1,$$

where $E_{\max}$ is the maximum value of the quasiparticle energy $E(p)$ (1.28). Therefore, only the "fermion" weight survives in the thermodynamic limit in (4.8) and (5.2). In the thermodynamic limit, the determinants of the $M \times M$ matrices in expressions (4.10) and (5.4) for these contributions become the Fredholm determinants of the corresponding integral operators. This is explained explicitly, e.g., in [5]. Thus, we obtain the following answers.

The correlator $G(m)$ (4.7) is given by

$$G(m) = \lim \frac{1}{Z} \mathrm{Sp}\left(e^{\alpha Q(m)} e^{-\beta \mathbf{H}}\right) = \det\left(\hat{I} + (e^\alpha - 1)\hat{V}\right). \quad (6.1)$$

On the right-hand side of the above formula, we have the Fredholm determinant of a linear integral operator acting on the functions $f(p)$ defined on the interval $-\pi \le p \le \pi$ by the rule

$$(\hat{V} f)(p) = \int_{-\pi}^{\pi} V(p, q) f(q)\, dq, \quad (6.2)$$

where $\widehat{I}$ means the identity operator on the interval and the kernel $V(p,q)$ (see (4.10), (2.17), (4.12)) has the form

$$V(p,q)=\frac{1}{2\pi}\frac{\sin\frac{m(p-q)}{2}}{\sin\frac{p-q}{2}}\omega_F(q), \tag{6.3}$$

where the weight $\omega_F(q)$ is given by the formula

$$\omega_F(q)=\frac{1}{2}\left(1-e^{i\theta(q)}\tanh\frac{\beta E(q)}{2}\right) \tag{6.4}$$

(the angle $\theta(p)$ and the quasiparticle energy $E(p)$ are defined in (1.27) and (1.28)).

The thermodynamic limit of the correlators (5.1)

$$G^{(ab)}(m)=\lim\frac{1}{Z}\mathrm{Sp}\,\left(\sigma^a_{m+1}\,\sigma^b_1\,e^{-\beta\mathbf{H}}\right) \tag{6.5}$$

can also be represented in terms of Fredholm determinants of linear integral operators on the interval $[-\pi,\pi]$. For $m>0$, we obtain from (5.4)–(5.6)

$$G^{(-+)}(m)=G^{(+-)}(m)=\frac{\partial}{\partial\alpha}\det\left(I-\widehat{W}+\alpha\widehat{C}\right)\big|_{\alpha=0}\quad(m>0) \tag{6.6}$$

$$G^{(++)}(m)=G^{(--)}(m)=\frac{\partial}{\partial\alpha}\det\left(I-\widehat{W}-i\alpha\widehat{S}\right)\big|_{\alpha=0}\quad(m>0). \tag{6.7}$$

The kernels of the operators $\widehat{W}$, $\widehat{C}$, and $\widehat{S}$ are

$$W(p,q)=-\frac{1}{\pi}e^{\frac{i(p-q)}{2}}\frac{\sin\frac{m(p-q)}{2}}{\sin\frac{p-q}{2}}\omega_F(q), \tag{6.8}$$

$$C(p,q)=\frac{1}{2\pi}\cos\frac{m(p+q)}{2}, \tag{6.9}$$

$$S(p,q)=\frac{1}{2\pi}\sin\frac{m(p+q)}{2}. \tag{6.10}$$

These answers differ only by the weight (5.30) from the answers for the isotropic XXO chain.

It should be noted that the formal answers for the time-dependent correlation functions have recently been obtained in [29]. In a forthcoming paper, we expect to give more explicit representations for the time-dependent correlators.

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Appendix A

Consider the set of canonical fermion operators

$$a_q,\ a_q^+\quad\left([a_p,\,a_q]_+=[a_p^+,\,a_q^+]_+=0,\ [a_p,\,a_q^+]_+=\delta_{pq}\right).$$

We denote by M the number of operators a (or a^+) in the set. We assume that the Fock vacuum exists and has the following properties:

$$a_q\,|\,0\,\rangle=0,\quad\langle\,0\,|\,a_q^+=0\ \text{ for all }\ q,\quad\langle\,0\,|\,0\,\rangle=1. \tag{A.1}$$

We introduce the coherent states [30]

$$|\,\xi\,\rangle\equiv|\,\xi_q\,\rangle=\exp\Big\{\sum_q a_q^+\,\xi_q\Big\}\,|\,0\,\rangle,\quad|\,\xi^*\,\rangle\equiv|\,\xi_q^*\,\rangle=\langle\,0\,|\,\exp\Big\{\sum_q \xi_q^*\,a_q\Big\}, \tag{A.2}$$

where the summation is taken over the entire set. The parameters ξ_q and ξ_q^+ (Grassmann algebra elements) anticommute between each other and with the operators a_q and a_q^+. It should be noted that the asterisk in ξ^* means only that the corresponding parameter is connected with a bra-vector. We do not consider involutions on the Grassmann algebra; the parameters ξ and ξ^* are entirely independent. The coherent states (A.2) are the eigenstates for the creation and annihilation operators, respectively:

$$a_p \mid \xi \rangle = \xi_p \mid \xi \rangle, \quad \langle \xi_p^* \mid a_p^+ = \langle \xi^* \mid \xi_p. \tag{A.3}$$

Using the commutation relations between a, a^+, ξ, and ξ^*, one can easily calculate the scalar product of two coherent states:

$$\langle \xi^* \mid \xi \rangle = \exp\Big\{\sum_q \xi_q^* \xi_q\Big\} \equiv \exp\{\xi^* \xi\}. \tag{A.4}$$

The trace of an operator $\mathbf{O}$ can be represented as an integral over the anticommuting variables of the matrix elements of the operator between the coherent states (see [30]):

$$\mathrm{Sp}\,\mathbf{O} = \int d\xi\, d\xi^* \exp\{\xi^* \xi\} \langle \xi^* \mid \mathbf{O} \mid \xi \rangle, \tag{A.5}$$

and the expansion for the identity operator is given by

$$\mathbf{1} = (-1)^M \int d\xi\, d\xi^* \exp\{-\xi^* \xi\} \mid \xi \rangle \langle \xi^* \mid \tag{A.6}$$

(we assume that the number M of sites is even, and the coefficient $(-1)^M$ in such formulas is omitted). We use the following notation:

$$d\xi \equiv \prod_q d\xi_q, \quad d\xi^* \equiv \prod_q d\xi_q^*.$$

If an operator $\mathbf{L}$ has the form

$$\mathbf{L} = \sum_{p,q} a_p^+ L_{pq} a_q \equiv a^+ L a,$$

where L is an $M \times M$ matrix (with entries L_{pq}) which can be diagonalized by a unitary matrix U,

$$L = U^+ D U, \quad U^+ U = U U^+ = I, \quad D = \mathrm{diag}\,(D_q),$$

then the matrix elements of the operator $\exp \mathbf{L}$ are given by

$$\langle \xi^* \mid \exp\{\mathbf{L}\} \mid \xi \rangle = \exp\Big\{\sum_{p,q} \xi_p^* (\exp\{L\})_{pq} \xi_q\Big\} \exp\{\xi^* \exp L \xi\}, \tag{A.7}$$

and

$$\mathrm{Sp}\,\exp\{\mathbf{L}\} = \mathrm{Sp}\,\exp\{a^+ L a\} = \det\,[I + \exp\{L\}] \tag{A.8}$$

(we have the trace of an operator on the left-hand side of the above formula and the determinant of an $M \times M$ matrix on the right-hand side).

Equation (A.8) follows from the well-known formula for the Gauss integral over anticommuting variables:

$$\int d\xi\, d\xi^* \exp\{\xi^* K \xi + \eta^* \xi + \xi^* \eta\} = \exp\{-\eta^* K^{-1} \eta\} \det K. \tag{A.9}$$

We also need the following formula, which is valid for an antisymmetric matrix A:

$$\int d\xi \exp\Big\{\frac{1}{2} \xi A \xi + \eta \xi\Big\} = \exp\Big\{\frac{1}{2} \eta A^{-1} \eta\Big\} \sqrt{\det A}, \quad A = -A^T. \tag{A.10}$$

Appendix B

In this section, we obtain the formulas for the time-dependent correlators of the isotropic XXO chain with finite number M of sites (it is assumed that M is even). We hope that the reader will appreciate the simplicity of the proof using integration over the Grassmannian variables (cf. [5, 26]). We begin with the calculation of the partition function Z. Following [23, 25], we represent Z in the form

$$Z = \mathrm{Sp}\,\exp\{-\beta\mathbf{H}\} = \frac{1}{2}(Z_F^+ + Z_F^- + Z_B^+ - Z_B^-), \tag{B.1}$$

where $\sigma_{m+1}^- \equiv 1/T$ is the inverse temperature and the contributions on the right-hand side are defined by the formulas

$$\begin{aligned} Z_F^\pm &= \mathrm{Sp}\,\exp\{-\beta\mathbf{H}^\pm\} = e^{\beta Mh/2}\det(I + J_\beta), \\ Z_B^\pm &= \mathrm{Sp}\,\left[(-1)^{\mathbf{N}}\exp\{-\beta\mathbf{H}^\pm\}\right] = e^{\beta Mh/2}\det(I - J_\beta), \end{aligned} \tag{B.2}$$

where J_β is a diagonal matrix,

$$J_\beta = \mathrm{diag}\,(e^{-\beta\varepsilon(p)}) \tag{B.3}$$

(we note that $p \in X^+$ for $Z_{F,B}^+$ and $p \in X^-$ for $Z_{F,B}^-$). To obtain these representations, we use the following relations:

$$(-1)^{\mathbf{N}} \mid \phi, \pm\rangle = \mid -\phi, \pm\rangle, \tag{B.4}$$

$$\exp\{-\beta\mathbf{H}^\pm \mid \phi, \pm\rangle = e^{\beta Mh/2} \mid J_\beta\phi, \pm\rangle \equiv e^{\beta Mh/2}\exp\Big\{\sum_{p\in X^\pm} a_p^+\, e^{-\beta\varepsilon(p)}\phi_p\Big\} \mid 0\rangle \tag{B.5}$$

and rewrite Z_B^+ as a Gauss integral:

$$Z_B^\pm = e^{\beta Mh/2}\int d\phi\, d\phi^*\exp\{\phi^*\,[I - J_\beta]\,\phi\} \tag{B.6}$$

(see (A.4), (A.5)).

Now we consider the simplest equal-time temperature correlator

$$G(m) \equiv \frac{1}{Z}\mathrm{Sp}\,\left[e^{\alpha\mathbf{Q}(m)}\,e^{-\beta\mathbf{H}}\right], \tag{B.7}$$

which can be represented in the following form [26]:

$$G(m) = \frac{1}{2Z}\left(Z_F^+\,G_F^+ + Z_F^-\,G_F^- + Z_B^+\,G_B^+ - Z_B^-\,G_B^-\right), \tag{B.8}$$

where

$$Z_F^\pm\,G_F^\pm = \mathrm{Sp}\,\left[e^{\alpha\mathbf{Q}(m)}\,e^{-\beta\mathbf{H}^\pm}\right],\quad Z_B^\pm\,G_B^\pm = \mathrm{Sp}\,\left[e^{\alpha\mathbf{Q}(m)}\,e^{-\beta\mathbf{H}^\pm}\,(-1)^{\mathbf{N}}\right]. \tag{B.9}$$

The contributions can be represented as determinants of $M \times M$ matrices:

$$G_F^\pm = \det\,(I + (e^\alpha - 1)\,Q^{(0)}(m)\,\Theta_F);\quad G_B^\pm = \det\,(I + (e^\alpha - 1)\,Q^{(0)}(m)\,\Theta_F), \tag{B.10}$$

where the diagonal matrices Θ_F and Θ_B of the Fermi and Bose weights have the form

$$\Theta_F = J_\beta[I + J_\beta]^{-1} = \mathrm{diag}\left[\Theta_F(p) \equiv \frac{1}{1 + e^{\beta\varepsilon(p)}}\right],\quad \Theta_B = J_\beta[I + J_\beta]^{-1} = \mathrm{diag}\left[\Theta_B(p) \equiv \frac{1}{e^{\beta\varepsilon(p)} - 1}\right], \tag{B.11}$$

and the matrix $Q^{(0)}(m)$ with entries $Q_{pq}^{(0)}(m)$ was defined in (2.17) ($p, q \in X^+$ for $G_{F,B}^+$ and $p, q \in X^-$ for $G_{F,B}^-$). To obtain these representations, one should rewrite, for example, the contribution of $Z_B^- G_B^-$ as

$$Z_B^-\,G_B^- = e^{\beta Mh/2}\int d\phi\, d\phi^*\, e^{\phi^*\phi}\langle\phi^*, - \mid e^{\alpha\mathbf{Q}(m)} \mid -J_\beta\phi, -\rangle,$$

use expression (2.18) for the matrix element under the integral sign, calculate the Gauss integral, perform the similarity transformation $Q(m) \to Q^{(0)}(m)$ (see (2.16), (2.17)), and factor out the coefficient $\det(I - J_\beta)$ from the determinant obtained.

Now we turn to the time-dependent correlation function of the local spins. As usual, we have

$$\sigma_m^\alpha(t) = e^{it\mathbf{H}}\,\sigma_m^\alpha\,e^{-it\mathbf{H}}.$$

In consequence of the translation invariance, the local spin correlators $\sigma_{m_2}^\pm(t_2)$ and $\sigma_{m_1}^\pm(t_1)$ depend only on the differences $m = m_2 - m_1$ and $t = t_2 - t_1$:

$$G^{(ab)}(m,t) = \frac{1}{Z}\mathrm{Sp}\left[e^{iHt}\,\sigma_{m+1}^a\,e^{-iHt}\,\sigma_1^b\,e^{-\beta H}\right] \quad (a,b = \pm). \tag{B.12}$$

Due to (2.21), the correlator can be represented as a sum of four contributions:

$$G^{(ab)}(m,t) = \frac{1}{2Z}\left(Z_F^+\,G_F^{(ab),+} + Z_F^-\,G^{(ab),-} + Z_B^+\,G^{(ab),+} - Z_B^-\,G^{(ab),-}\right), \tag{B.13}$$

where

$$\begin{aligned} Z_F^\pm\,G_F^{(ab),\pm} &= \mathrm{Sp}\left[e^{i\mathbf{H}^\pm t}\,\sigma_{m+1}^a\,e^{-i\mathbf{H}^\mp t}\,\sigma_1^b\,e^{-\beta\mathbf{H}^\pm}\right], \\ Z_B^\pm\,G_B^{(ab),\pm} &= \mathrm{Sp}\left[e^{i\mathbf{H}^\pm t}\,\sigma_{m+1}^a\,e^{-i\mathbf{H}^\mp t}\,\sigma_1^b\,e^{-\beta\mathbf{H}^\pm}\,(-1)^{\mathbf{N}}\right]. \end{aligned} \tag{B.14}$$

We begin with the correlator $G^{(-+)}(m,t)$. We have

$$Z_F^+\,G^{(-+),+} = e^{\beta Mh/2}\int d\phi\,d\phi^*\,d\psi\,d\psi^*\,e^{\phi^*\phi - \psi^*\psi}\langle\,\phi^*\,J(0,t),+\mid\sigma_{m+1}^-\mid J(0,-t)\psi,-\rangle\langle\,\psi^*,-\sigma_1^+\mid J_\beta\phi,+\rangle,$$

where

$$J(m,t) = \mathrm{diag}\left(e^{-ipm + i\varepsilon(p)t}\right). \tag{B.15}$$

Now we use the expressions for the matrix elements (2.22) and perform the integration over ψ, ψ^* and then over ϕ, ϕ^*,

$$\begin{aligned} Z_F^+\,G_F^{(-+),+} &= e^{\beta Mh/2}\frac{\partial}{\partial\alpha}\int d\phi\,d\phi^*\,d\psi\,d\psi^*\,e^{\phi^*\phi - \psi^*\psi}e^{\phi^*\,L(m,t)\psi + \psi^*\,L(0)J_\beta\phi + \alpha\phi^*\,R\,J_\beta\,\phi}\Big|_{\alpha=0} \\ &= e^{\beta Mh/2}\frac{\partial}{\partial\alpha}\det\left[I + L(m,t)\,L(0)\,J_\beta + \alpha\,R\,J_\beta\right]\Big|_{\alpha=0}, \end{aligned}$$

where

$$L(m,t) = J(0,t)\,L(m)\,J(0,-t)$$

(for $L_{p,q}(m,t)$, we have $p \in X^+$ and $q \in X^-$), and R is a matrix of rank 1 with equal columns:

$$R_{p_1p_2} = \frac{1}{M}e^{-ip_1m + i\varepsilon(p_1)t}, \quad p_1,\,p_2 \in X^+.$$

After the similarity transformation with the diagonal matrix $J(-\frac{m}{2}, -\frac{t}{2})$, we obtain

$$Z_F^+\,G_F^{(-+),+} = e^{\beta Mh/2}\frac{\partial}{\partial\alpha}\det\left[I + \widetilde{L}\,J_\beta + \alpha\,\widetilde{R}\,J_\beta\right]\Big|_{\alpha=0}, \tag{B.16}$$

where the entries of $\tilde{L}$ and $\tilde{R}$ have the form

$$\widetilde{L}_{p_1p_2} = e^{imp_1/2 - it\varepsilon(p_1)/2}\sum_q L_{p_1q}(m,t)L_{qp_2}(0)e^{-imp_2/2 + it\varepsilon(p_2)/2} \tag{B.17}$$

$$\widetilde{R}_{p_1p_2} = \frac{1}{M}\, e^{-imp_1/2+it\varepsilon(p_1)/2}\, e^{-imp_2/2+it\varepsilon(p_2)/2} \quad (\text{here} \quad p_1,\, p_2 \in X^+, \quad q \in X^-). \tag{B.18}$$

To represent these matrices in a more convenient form, we introduce the following functions (we follow [5], slightly changing the notation; recall that $\varepsilon(q) = h - \cos q$):

$$g(m,t) = \frac{1}{M}\sum_q e^{imq+it\cos q}, \tag{B.19}$$

$$e(m,t,p) = \frac{1}{M}\sum_q \frac{e^{imq+it\cos q}}{\tan\frac{q-p}{2}}, \tag{B.20}$$

$$d(m,t,p) = \frac{1}{M}\sum_q \frac{e^{imq+it\cos q} - e^{imp+it\cos p}}{\sin^2\frac{q-p}{2}}, \tag{B.21}$$

$$e_-(m,t,p) = e^{-i\frac{m}{2}p - i\frac{t}{2}\cos p}, \tag{B.22}$$

$$e_+(m,t,p) = e_-(m,t,p)\, e(m,t,p). \tag{B.23}$$

All these functions are of order $O(1)$ as $M \to \infty$. It is also convenient to introduce four one-dimensional projectors Π^{++}, Π^{+-}, Π^{-+}, and Π^{--}, which are $M \times M$ matrices with the entries

$$\Pi^{ab}_{p_1p_2} = e_a(m,t,p_1)\, e_b(m,t,p_2) \quad (a,b) = +,-). \tag{B.24}$$

Using identities

$$\sum_q \frac{1}{\sin^2\frac{q-p}{2}} = M^2 \quad (q \in X^-, \quad p \in X^+), \tag{B.25}$$

$$\cot(x-u)\cot(x-v) = \cot(u-v)[\cot(x-u) - \cot(x-v)] - 1, \tag{B.26}$$

we can represent the matrix $\tilde{L}$ in the following form:

$$\widetilde{L} = I + \frac{1}{M}\left[S + i\Pi^{+-} - i\Pi^{-+}\right], \tag{B.27}$$

where the diagonal and nondiagonal entries of the matrix S have the form

$$S_{pp} = d(p)\, e^{-imp-it\cos p}, \tag{B.28}$$

$$S_{p_1p_2} = \frac{e_+(p_1)\, e_-(p_2) - e_-(p_1)\, e_+(p_2)}{\tan\frac{p_1-p_2}{2}}. \tag{B.29}$$

The matrix $\tilde{R}$ is a projector:

$$\widetilde{R} = \frac{e^{ith}}{M}\Pi^{--}. \tag{B.30}$$

(For brevity, the arguments m and t, which are the same for all the functions d, e, and $e_\pm$, are omitted).

The other contributions can be calculated similarly. We present only the result. We have

$$\begin{aligned} G_F^{(-+),\pm} &= e^{ith}\frac{\partial}{\partial\alpha}\det\left[I + \frac{1}{M}(S + i\Pi^{+-} - i\Pi^{-+} + \alpha\Pi^{--})\Theta_F\right]\Big|_{\alpha=0}, \\ G_B^{(-+),\pm} &= e^{ith}\frac{\partial}{\partial\alpha}\det\left[I - \frac{1}{M}(S + i\Pi^{+-} - i\Pi^{-+} + \alpha\Pi^{--})\Theta_B\right]\Big|_{\alpha=0}. \end{aligned} \tag{B.31}$$

Although both contributions of $G_F^{(-+),\pm}$ (as well as of $G_B^{(-+),\pm}$) are written in the same form, they are, in fact, different. It is necessary to take into account that

$$p,\, p_1,\, p_2 \in X^+, \quad q \in X^- \quad \text{for} \quad G_{F,B}^{(-+),+}$$

and

$$p, p_1, p_2 \in X^-, \quad q \in X^+ \quad \text{for} \quad G_{F,B}^{(-+),-}$$

in all formulas (B.19)–(B.24), (B.28), and (B.29) defining the functions g, e, d, $e_\pm$, and the entries of the matrices S and Π^{ab}.

Similarly, one can calculate the correlator $G^{(+-)}$. The corresponding contributions are

$$G_F^{(+-),\pm} = e^{-ith}\Big[g(m,t)+\frac{\partial}{\partial\alpha}\Big]\det\Big[I+\frac{1}{M}(S+i\Pi^{+-}-i\Pi^{-+}-\alpha\{\Pi^{++}+ig\Pi^{+-}-ig\Pi^{-+}+g^2\Pi^{--}\})\Theta_F\Big]\Big|_{\alpha=0}.$$

(Once again, one should take into account the distinction between the contributions of $G_{F,B}^{(+-),+}$ and $G_{F,B}^{(+-),-}$ explained above).

It should be noted that the results obtained here coincide with the representations obtained by another method (which requires very complicated calculations) in [5, 26]. More precisely, our expressions can be obtained by choosing arbitrary constants $c_1 = i$, $c_2 = -i$ in formulas (A.1)–(A.14) of Appendix A of [28].

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